MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, ADJUSTABLE, NEGATIVE
VOLTAGE REGULATORS, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/118, dated 31 March 1980, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1: Delete "Three product assurance classes," and substitute "Two product assurance classes".

PAGE 3

TABLE I, V_{OUT}, (RECOV), load current column: Delete "C_L = 2.0 µF" as substituted in amendment 1 and reinstate "C_L = 20 µF".

TABLE I, V_{START}, load current column: Delete "C_L = 2.0 µF" as substituted in amendment 1 and reinstate "C_L = 20 µF".

PAGE 4

TABLE I, V_{OUT}, (RECOV), load current column: Delete "C_L = 2.0 µF" as substituted in amendment 1 and reinstate "C_L = 20 µF".

TABLE I, V_{START}, load current column: Delete "C_L = 2.0 µF" as substituted in amendment 1 and reinstate "C_L = 20 µF".

PAGE 5

* TABLE I, V_{RTH}, load current column and limits columns: Delete "I_L = 750 mA" and substitute "I_L = 500 mA".

* TABLE I, V_{OUT} (RECOV), load current column: Delete "C_L = 20 µF" and substitute "C_L = 10 µF".

* TABLE I, V_{START}, load current column: Delete "C_L = 20 µF" and substitute "C_L = 10 µF".

* TABLE I, $\Delta V_{IN}/\Delta V_{OUT}$, other column: Delete "4/" and substitute "6/".

* TABLE I, $\Delta V_{OUT}/\Delta V_{IN}$, input voltage and load current columns: Delete "V_{IN} = -1.0 V" and "I_L = 10 mA" and substitute "ΔV_{IN} = -1.0 V" and "I_L = 50 mA", respectively.

PAGE 6

TABLE I, I_{OS2}, maximum limits column: Delete "0.8" and substitute "1.0".

* TABLE I, V_{OUT} (RECOV), load current column: Delete "C_L = 20 µF" and substitute "C_L = 10 µF".

* TABLE I, V_{START}, load current column: Delete "C_L = 20 µF" and substitute "C_L = 10 µF".

* TABLE I, $\Delta V_{OUT}/\Delta V_{IN}$, load current and maximum limits columns: Delete "I_L = 10 mA" "40" and substitute "I_L = 100 mA", "80", respectively.

* TABLE I, $\Delta V_{IN}/\Delta V_{OUT}$, other column: Delete "4/" and substitute "6/".

* TABLE I, notes: Add "6/ The meter for e_i and e_o shall have a minimum bandwidth from 300 Hz to 10 kHz and shall measure true rms voltage.".

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:25Z
Check the source to verify that this is the current version before use.

PAGE 7

* 3.3.1: Add the following,

"Type Y

copper core - 24.75 per cent maximum
clad with Alloy 52 - 75.25 per cent maximum "

3.5, delete and substitute:

"3.5 Electrical test requirements. Electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III."

PAGE 12

FIGURE 4, at the top of the figure: Delete " see note 7 " and substitute " see note 8 ".

* FIGURE 4, symbol A, maximum dimension in inches and millimeters: Delete ".450" and "11.43" and substitute ".352" and "8.94".

PAGE 17

FIGURE 10, 1.0 µF capacitor: Change the polarity.

* FIGURE 10, note 2: Delete "4 hours" and substitute "30 minutes".

FIGURE 11, capacitors C_f and C_L: Change the polarity.

PAGE 18

* FIGURE 12, Device table, C_f and C_L for device types 03 and 04: Delete "1.0 uF" and substitute "10.0 uF". (2 places)

PAGE 24

* FIGURE 15, Device table, for device type 03: Delete "ΔV_{IN} = -3.0 V" and "R_L = 120Ω" and substitute "ΔV_{IN} = -1.0 V" and "R_L = 25Ω".

* FIGURE 15, Device table, for device type 04: Delete "ΔV_{IN} = -3.0 V" and "R_L = 120Ω" and substitute "ΔV_{IN} = -1.0 V" and "R_L = 12Ω".

PAGE 33

* TABLE III, test number 10, test conditions column I_L and limits columns: Delete "750" and substitute "500".

*TABLE III, test number 10, equations column: Delete "V_{RTH} = Δt E7" and substitute "V_{RTH} = E8".

PAGE 35

* TABLE III, test number 71, test conditions, input voltage, load current and equation columns: Delete "V_{pulse} = -3.0 V", "I_L = 10 mA", "E5/3" and substitute "V_{pulse} = -1.0 V", "I_L = 50 mA", "E59/1", respectively.

PAGE 36

TABLE III, test number 17, maximum limits column: Delete "0.8" and substitute "1.0".

PAGE 37

TABLE III, test number 38, maximum limits column: Delete "0.8" and substitute "1.0".

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:25Z
Check the source to verify that this is the current version before use.

PAGE 38

TABLE III, test number 60, maximum limits column: Delete "0.8" and substitute "1.0".

* TABLE III, test number 68, test conditions, input voltage, load current equation and maximum limits columns: Delete "V_{pulse} = -3.0 V", "I_L = 10 mA", "E59/3", "40", and substitute "V_{pulse} = -1.0 V", "I_L = 100 mA", "E59/1", "80", respectively.

PAGE 39

3.6: Delete second sentence in its entirety.

4.2e, delete and substitute the following: "e. Percent Defective Allowable (PDA). The PDA for class S and class B devices shall be as specified in MIL-M-38510".

TABLE II: Delete class C device column in its entirety.

PAGE 40

4.4.3a, delete and substitute the following: "a. End-point electrical parameters shall be as specified in table IV herein. Delta limits shall apply to subgroup 1 of group C inspection for class B devices."

4.4.3b: Delete "and C".

4.6: Delete in its entirety.

PAGE 41

TABLE IV, device types 03 and 04, symbol V_{OUT}, limits column: Delete "-1.30" and "-1.20" and substitute "-1.275" and "-1.225" respectively."

The margins of this amendment are marked with asterisks or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Air Force - 17
Army - ER
Navy - EC
NASA - NA

Review activities:
Air Force - 11, 19, 85, 99
Army - AR, MI
Navy - OS, SH
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1069)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:25Z
Check the source to verify that this is the current version before use.